

Supplementary Appendix for “Unveiling Spectral Misalignment: Generalizable Region Representation via Urban Pattern Consensus”

This supplementary appendix is organized as follows. Appendix A gives detailed proofs of the theoretical results. Appendix B provides the pretraining algorithm. Appendix C describes the data sources and view-native graph construction. Appendix D summarizes downstream task construction, the linear-probe evaluation protocol, and the compared baselines.

APPENDIX A PROOFS OF THEORETICAL RESULTS

A. Common Notation and Assumptions

We present the proofs under the same simplified single-view abstraction used in the main text. The arguments apply independently to each SURE view, including POI, land use, and mobility. For city c , let a region-level representation be decomposed as $H^{(c)} = C + B + N^{(c)}$, where C denotes transferable commonality shared across cities, B denotes task-relevant local specificity, and $N^{(c)}$ denotes city-specific spectral nuisance. All random variables are assumed to be centered. If the variables are not centered, the same derivations hold after centering or by including an intercept term in the predictor.

Unless otherwise stated, the components are pairwise uncorrelated, i.e., $\text{Cov}(C, B) = 0$, $\text{Cov}(C, N^{(c)}) = 0$, and $\text{Cov}(B, N^{(c)}) = 0$. The downstream target is generated by $y = \theta_C^\top C + \theta_B^\top B + \xi$, where ξ is zero-mean noise uncorrelated with C , B , and $N^{(c)}$. We write $\Sigma_C = \text{Cov}(C)$, $\Sigma_B = \text{Cov}(B)$, and $\Sigma_N^{(c)} = \text{Cov}(N^{(c)})$.

B. Proof of Theorem 1

Proof. Consider a linear predictor $\hat{y} = w^\top H^{(c)}$. The population squared loss in city c is

$$\mathcal{R}_c(w) = \mathbb{E} \left[(y - w^\top H^{(c)})^2 \right].$$

Expanding the loss gives

$$\mathcal{R}_c(w) = \mathbb{E}[y^2] - 2w^\top \mathbb{E}[H^{(c)}y] + w^\top \mathbb{E}[H^{(c)}(H^{(c)})^\top]w.$$

Taking the derivative with respect to w and setting it to zero yields the normal equation

$$\mathbb{E}[H^{(c)}(H^{(c)})^\top]w_c^* = \mathbb{E}[H^{(c)}y].$$

Because all variables are centered, this is equivalent to

$$\text{Cov}(H^{(c)})w_c^* = \text{Cov}(H^{(c)}, y).$$

We first compute the representation covariance. Since $H^{(c)} = C + B + N^{(c)}$, pairwise uncorrelatedness gives

$$\begin{aligned} \text{Cov}(H^{(c)}) &= \text{Cov}(C + B + N^{(c)}) \\ &= \Sigma_C + \Sigma_B + \Sigma_N^{(c)}. \end{aligned}$$

Next, using $y = \theta_C^\top C + \theta_B^\top B + \xi$, we have

$$\begin{aligned} \text{Cov}(H^{(c)}, y) &= \text{Cov} \left(C + B + N^{(c)}, \theta_C^\top C + \theta_B^\top B + \xi \right) \\ &= \text{Cov}(C, \theta_C^\top C) + \text{Cov}(B, \theta_B^\top B), \end{aligned}$$

because all cross-covariance terms involving different components or ξ are zero. For centered vectors,

$$\text{Cov}(C, \theta_C^\top C) = \Sigma_C \theta_C, \quad \text{Cov}(B, \theta_B^\top B) = \Sigma_B \theta_B.$$

Therefore,

$$\text{Cov}(H^{(c)}, y) = \Sigma_C \theta_C + \Sigma_B \theta_B.$$

Substituting these two expressions into the normal equation gives

$$(\Sigma_C + \Sigma_B + \Sigma_N^{(c)})w_c^* = \Sigma_C \theta_C + \Sigma_B \theta_B.$$

Under the nonsingularity condition in Theorem 1,

$$w_c^* = (\Sigma_C + \Sigma_B + \Sigma_N^{(c)})^{-1}(\Sigma_C \theta_C + \Sigma_B \theta_B),$$

which is the expression stated in Theorem 1. If the covariance matrix is singular, the same normal-equation argument still yields a least-squares solution using the Moore-Penrose pseudoinverse or an arbitrarily small ridge regularizer; the city-dependence argument below applies analogously to the regularized optimum.

It remains to show why different nuisance covariances generally lead to different optima. Let

$$v = \Sigma_C \theta_C + \Sigma_B \theta_B.$$

For two cities a and b ,

$$w_a^* = (\Sigma_C + \Sigma_B + \Sigma_N^{(a)})^{-1}v, \quad w_b^* = (\Sigma_C + \Sigma_B + \Sigma_N^{(b)})^{-1}v.$$

If the two optima were equal to the same vector w , then

$$(\Sigma_C + \Sigma_B + \Sigma_N^{(a)})w = v, \quad (\Sigma_C + \Sigma_B + \Sigma_N^{(b)})w = v.$$

Subtracting the two equations gives

$$(\Sigma_N^{(a)} - \Sigma_N^{(b)})w = 0.$$

Thus, a shared optimum requires w to lie in the null space of $\Sigma_N^{(a)} - \Sigma_N^{(b)}$. When $\Sigma_N^{(a)} \neq \Sigma_N^{(b)}$, this is a special alignment condition rather than a generic property. Hence the population optima are generally city-dependent.

The same argument also explains the downstream risk caused by transferring a city-specific optimum. For city b , the quadratic risk can be written as

$$\mathcal{R}_b(w) = \mathcal{R}_b(w_b^*) + (w - w_b^*)^\top \Sigma_H^{(b)} (w - w_b^*),$$

where $\Sigma_H^{(b)} = \Sigma_C + \Sigma_B + \Sigma_N^{(b)}$. Therefore,

$$\mathcal{R}_b(w_a^*) - \mathcal{R}_b(w_b^*) = (w_a^* - w_b^*)^\top \Sigma_H^{(b)} (w_a^* - w_b^*) \geq 0.$$

The inequality is strict whenever $w_a^* \neq w_b^*$ and $\Sigma_H^{(b)}$ is positive definite along the difference direction. This completes the proof. $\square$

C. Proof of Theorem 2

Proof. Recall that $H_\alpha^{(c)} = C + \alpha R^{(c)}$, $R^{(c)} = B + N^{(c)}$, and the task-relevant latent factor is $T = C + \nu B$. Therefore,

$$\begin{aligned} H_\alpha^{(c)} &= C + \alpha B + \alpha N^{(c)}, \\ T - H_\alpha^{(c)} &= (\nu - \alpha)B - \alpha N^{(c)}. \end{aligned}$$

Because B and $N^{(c)}$ have zero means and zero cross-correlation, $\mathbb{E}[B^\top N^{(c)}] = 0$,

$$\begin{aligned} \mathbb{E}\|T - H_\alpha^{(c)}\|_2^2 &= \mathbb{E}\|(\nu - \alpha)B - \alpha N^{(c)}\|_2^2 \\ &= (\nu - \alpha)^2 \mathbb{E}\|B\|_2^2 + \alpha^2 \mathbb{E}\|N^{(c)}\|_2^2. \end{aligned}$$

Let $m_B = \mathbb{E}\|B\|_2^2$ measure useful specificity and let $m_N = \mathbb{E}\|N^{(c)}\|_2^2$ measure nuisance strength. With $\mu \geq 0$ and $\Delta_N^2 \geq 0$, adding the local-variation shift penalty $\mu\alpha^2\Delta_N^2$ gives the objective

$$\mathcal{J}(\alpha) = (\nu - \alpha)^2 m_B + \alpha^2 m_N + \mu\alpha^2 \Delta_N^2.$$

Expanding the quadratic terms,

$$\mathcal{J}(\alpha) = \nu^2 m_B - 2\nu m_B \alpha + (m_B + m_N + \mu\Delta_N^2)\alpha^2.$$

The first derivative is

$$\frac{d\mathcal{J}}{d\alpha} = -2\nu m_B + 2(m_B + m_N + \mu\Delta_N^2)\alpha,$$

and the second derivative is

$$\frac{d^2\mathcal{J}}{d\alpha^2} = 2(m_B + m_N + \mu\Delta_N^2) \geq 0.$$

Thus $\mathcal{J}$ is convex in α . If $m_B + m_N + \mu\Delta_N^2 > 0$, setting the first derivative to zero gives

$$(m_B + m_N + \mu\Delta_N^2)\alpha^* = \nu m_B,$$

and hence

$$\alpha^* = \frac{\nu m_B}{m_B + m_N + \mu\Delta_N^2},$$

which is the optimizer stated in Theorem 2. If $m_B + m_N + \mu\Delta_N^2 = 0$, then $m_B = m_N = 0$ and $\mu\Delta_N^2 = 0$, so $\mathcal{J}(\alpha) = 0$ for all $\alpha \in [0, 1]$. In this degenerate case, every α is optimal and there is no useful local-variation trade-off.

Algorithm 1 SURE Pretraining

Require: City collection $\mathcal{C}$, global epochs E , local repeats L_{loc}
Ensure: Pretrained city-shared encoder f_θ

- 1: Initialize model parameters θ and the shared urban pattern prototype bank.
- 2: **for** $e = 1, \dots, E$ **do**
- 3: **for** each city $c \in \mathcal{C}$ **do**
- 4: **for** $s = 1, \dots, L_{\text{loc}}$ **do**
- 5: **for** each view-native graph $v \in \mathcal{V}$ **do**
- 6: Project features $H_v^{(c)} \leftarrow \phi_v(X_v^{(c)})$.
- 7: Extract low-frequency shared patterns $H_v^{(c), \text{sp}}$ with learnable graph spectral decomposition; set high-frequency local variations $H_v^{(c), \text{lv}} \leftarrow H_v^{(c)} - H_v^{(c), \text{sp}}$.
- 8: **end for**
- 9: Compute urban pattern assignment $Q^{(c)}$, urban pattern states $Y^{(c)}$, and consensus representation $C^{(c)}$.
- 10: Select high-frequency representation $R^{(c)}$ with consensus-guided local-variation attention.
- 11: Fuse $C^{(c)}$ and $R^{(c)}$ with an adaptive spectral gate to obtain $\tilde{Z}^{(c)}$.
- 12: Refine embeddings $Z^{(c)}$ with the spatial GCN.
- 13: Compute $\mathcal{L}_{\text{gr}}^{(c)}$, $\mathcal{L}_{\text{cv}}^{(c)}$, $\mathcal{L}_{\text{geo}}^{(c)}$, and $\mathcal{L}_{\text{bal}}^{(c)}$.
- 14: Combine them into the city-level pretraining objective $\mathcal{L}^{(c)}$.
- 15: Update θ with AdamW using $\nabla_\theta \mathcal{L}^{(c)}$.
- 16: **end for**
- 17: **end for**
- 18: **end for**
- 19: **return** f_θ .

Finally, since $m_B \geq 0$, $m_N \geq 0$, $\mu \geq 0$, and $\Delta_N^2 \geq 0$,

$$m_B + m_N + \mu\Delta_N^2 \geq m_B.$$

For $\nu \in [0, 1]$, this implies

$$0 \leq \alpha^* = \nu \frac{m_B}{m_B + m_N + \mu\Delta_N^2} \leq \nu \leq 1.$$

This proves the stated optimizer and its bound. $\square$

APPENDIX B SURE PRETRAINING ALGORITHM

APPENDIX C DATA AND GRAPH CONSTRUCTION

This section describes the tract-level input views used for SURE pretraining and the graph relations constructed from them. SURE is pretrained on three views: POI, land use, and human mobility. All view data are aligned by the 11-digit census-tract identifier GEOID20, so that the same tract order is shared by the view-specific feature matrices and graph relations.

A. POI and Land Use Views

POI and land use features are derived from Overture Maps. POIs are obtained from the Places theme and spatially joined to census tracts. Each place is represented by a semantic token derived from its Overture `basic_category`; tract-level POI features are then computed as category-wise counts, yielding 208 POI dimensions. Land use entities are obtained from the Base theme's LandUse layer. For each tract, the area-overlap

TABLE I
OVERTURE FEATURE FAMILIES RELATED TO POI, LAND COVER, AND LAND USE.

Family	Count	Role in this paper
POI	208	Semantic urban features describing tract-level service and facility composition.
Land cover	6	Physical surface categories used to build the dominant land-cover prediction task.
Land use	21	Functional land-use features used as heterogeneous urban signals.

ratio between the tract polygon and each land-use category is computed, yielding 21 land-use dimensions. Missing feature values are filled with zeros after the spatial aggregation.

The Overture feature matrix also contains land-cover features, which are used only to construct the downstream land-cover prediction task rather than as a pretraining view. Table I summarizes the Overture feature families used in this paper and their roles.

B. Human Mobility View

Human mobility is constructed from the LEHD Origin-Destination Employment Statistics (LODES) released by the U.S. Census Bureau. The main origin-destination records for all jobs (JT00) are used, where `h_geocode` denotes the home census block, `w_geocode` denotes the workplace census block, and `S000` records the total commuting flow count between the two blocks. The 2023 release is used when available; otherwise, the latest available state-level release is used.

To align the block-level OD records with census-tract regions, the LODES crosswalk maps each 2020 census block `tabblk2020` to its 11-digit tract identifier. Each OD record is then converted from a block-level pair (`h_geocode`, `w_geocode`) to a tract-level pair (home tract, work tract). Flows with both endpoints mapped to the selected tract set are retained and summed by tract pair. The resulting tract-level OD matrix M uses the aggregated `S000` value as the edge weight. When constructing county-level graph bundles, only OD rows whose home and workplace blocks both fall inside the county’s tract set are retained. The global bundle keeps flows whose endpoints fall in the union of all selected tracts, thereby preserving cross-county commuting links.

The stored mobility graph is symmetrized for representation learning: a commuting flow from tract i to tract j contributes weighted edges in both directions, while self-flows are kept once. The final sparse graph is saved together with the tract ordering and block-to-tract mapping. From this tract-level mobility matrix, the mobility feature matrix used in the main paper is derived, including flow volume, entropy, self-flow ratio, flow balance, and mobility centrality statistics.

C. View-Native Graph Construction

For POI and land use, the tract feature matrices are used to compute pairwise cosine similarity matrices. The POI similarity matrix is computed from the 208-dimensional POI count vectors, and the land-use similarity matrix is computed from the 21-dimensional land-use area-ratio vectors. These similarities define the corresponding view-native relations, from which each tract is connected to its top- k most similar semantic neighbors during graph construction. The mobility view is built from the LODES tract-level OD graph described above and can be sparsified by retaining the strongest outgoing flow edges for each tract. Thus, the three pretraining views: service composition similarity for POI, functional land-use similarity for land use, and commuting connectivity for human mobility.

APPENDIX D EVALUATION DETAILS

All downstream labels are constructed at the census-tract level and aligned with region embeddings by the 11-digit tract identifier GEOID20. The evaluation suite contains six tasks covering socioeconomic estimation, socioeconomic classification, employment-risk prediction, built-environment classification, public-safety classification, and multi-label carbon-sector prediction.

A. Downstream Task Construction

Table II summarizes the data source, target definition, and prediction type for each task.

B. Task-Specific Label Details

The definitions of population density, income, employment, and land cover are fully specified in Table II. Safety and carbon require additional processing details, which are provided below.

a) *Safety grades.*: The CrimeGrade-style conversion is percentile-based rather than based on fixed crime-rate thresholds. Within each city, tracts with positive crime rates are ranked so that lower crime rates receive higher safety percentiles. The safety percentile interval $[0, 100]$ is divided into 13 equal-width bins with step size $100/13$. The grade id is computed as the bin index, clipped to $[0, 12]$, and mapped to the ordered letter grades $\{F, D^-, D, D^+, C^-, C, C^+, B^-, B, B^+, A^-, A, A^+\}$. Thus, the lowest safety-percentile bin corresponds to F and the highest bin corresponds to A+. Tracts with nonpositive or missing crime rates do not determine the percentile cutoffs and are removed when generating the final safety targets.

b) *Carbon sectors.*: The tract-level carbon labels use the five 2023 City Health Dashboard sector files. The sector files are joined by GEOID20, only tracts with strictly positive total emissions are retained, and sector shares are normalized to sum to one before thresholding. An auxiliary EDGAR-based CO₂ processing pipeline is maintained for data auditing. This auxiliary pipeline starts from EDGAR_2025_GHG monthly

TABLE II
CONSTRUCTION OF DOWNSTREAM TASKS.

Task	Source	Target construction	Type
Population density	ACS 2023 and tract land area	The target is population density rather than raw population. For tract i , it is computed as the ACS population divided by tract land area and measured as people per 1000 m ² .	Regression
Income	ACS 2023 median household income	Tract-level median household income is discretized into seven ordered, merged B19001-style brackets: < \$25k, \$25k–\$49,999, \$50k–\$74,999, \$75k–\$99,999, \$100k–\$149,999, \$150k–\$199,999, and ≥ \$200k.	7-class classification
Employment	ACS 2023 employment status	A high-unemployment label is constructed from ACS employment counts. For each tract, unemployment rate is computed as unemployed population divided by civilian labor force. The overall unemployment rate is estimated over the benchmark collection, and a tract is labeled positive if its unemployment rate is at least 1.2 times this overall rate; otherwise it is labeled negative.	Binary classification
Land cover	Overture Maps tract feature matrix	Each tract is assigned its dominant land-cover category among barren, crop, forest, grass, urban, and wetland. Tracts whose dominant land-cover value, i.e., the largest land-cover feature value, is below 0.1 are removed.	6-class classification
Safety	Public city crime records and ACS population	Crime incidents are spatially joined to census tracts using their coordinates. Incidents in each tract are counted and normalized by ACS population to obtain the crime rate per 100,000 residents, which is then converted into CrimeGrade-style safety levels. The labels range from 0 to 12, corresponding to F through A+.	13-class classification
Carbon	City Health Dashboard 2023 tract-level CO ₂ sector metrics	For each tract, CO ₂ emissions are organized over five sectors: commercial buildings, electricity production, industrial buildings, transportation, and residential buildings. Each sector’s share of the tract total is computed, and the sector label is set to positive if the share is at least 0.20, yielding a multi-hot target vector.	5-label multi-label classification

sector-specific CO₂ NetCDF gridmaps, clips each source to county boundaries, and computes area-weighted county-level totals and densities from pixel–county overlaps. Its processed records are indexed by county, sector, year, and month, and are therefore distinct from the tract-level carbon-sector labels used in the downstream evaluation.

C. Linear-Probe Evaluation

After frozen region embeddings are obtained, they are evaluated with linear probes. This protocol tests whether task-relevant information is linearly accessible from the learned embeddings rather than absorbed by a highly expressive downstream predictor. For each fold, invalid embedding values are set to zero, and a standardization transform is fitted on the training split and applied to the test split. All reported experiments use three-fold cross-validation. Regression tasks use shuffled K -fold splits, while classification tasks use stratified splits when class counts permit and otherwise fall back to shuffled K -fold splits. Metrics are computed from out-of-fold predictions, with fold-level standard deviations recorded for uncertainty.

Table III summarizes the probe and metrics used for each task, where ACC denotes accuracy and mAP denotes mean average precision. Population-density regression uses ridge

regression on the $\log(1 + y)$ -transformed target and maps predictions back to the original scale before evaluation. The ridge regularization strength is set to 1000 when the training fold has fewer than 50 samples and 100 otherwise. Classification probes use balanced logistic regression with the L-BFGS solver. For carbon prediction, one binary logistic classifier is trained for each emission sector, and predicted sector probabilities are thresholded at 0.5.

D. Compared Baselines

The compared methods are representative urban region representation learning approaches that cover multi-view graph learning, multi-graph mobility modeling, heterogeneous relation modeling, attentive fusion, contrastive prediction, graph-based pretraining and prompting, flexible region representation, and multi-view joint contrastive learning. For a fair comparison, the learned embeddings from each baseline are evaluated with the same linear-probe protocol described above.

- **MVURE** learns urban region embeddings by jointly modeling multiple view-specific urban graphs and their complementary relations [1].
- **MGFN** builds multiple mobility graphs from human mobility patterns and fuses them with intra-pattern and inter-pattern message passing [2].

TABLE III
LINEAR-PROBE EVALUATION PROTOCOL FOR DOWNSTREAM TASKS.

Task	Probe	Metrics
Population density	Ridge regression with log-target transform	MAE, RMSE
Income	Balanced multiclass logistic regression	ACC, Macro-F1
Employment	Balanced binary logistic regression	ACC, F1
Land cover	Balanced multiclass logistic regression	ACC, Macro-F1
Safety	Balanced multiclass logistic regression	ACC, Macro-F1
Carbon	Five one-vs-rest balanced logistic regressions	AUROC, mAP

- **HREP** models heterogeneous urban relations and uses prompt learning to adapt region embeddings to downstream tasks [3].
- **HAFusion** learns view-specific region embeddings and adaptively fuses heterogeneous urban signals with attention mechanisms [4].
- **ReCP** uses multi-view contrastive prediction to align complementary urban views and improve region embedding quality [5].
- **GURPP** introduces graph-based urban region pretraining with prompting to improve downstream transfer [6].
- **FlexiReg** learns urban representations under flexible region definitions and supports downstream adaptation across spatial settings [7].
- **MVJC** combines multi-view joint learning with contrastive objectives to integrate heterogeneous urban signals [8].

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